E9: 309 ADL 23-11-2020

http://leap.ee.iisc.ac.in/sriram/teaching/ADL2020/

Housekeeping

- ✴️ Mid-term exam
  - ➡️ December 5th (Saturday) [Topics covered up to Dec 2nd]
  - ➡️ Mode of exam
    - ✓ Time to respond - 3 hours
      - ◉ Exam paper uploaded in Teams Channel and response (photo-scanned and uploaded in your folder).
      - ◉ Open book, open notes
        - ★ Strictly no online communication or help sought.
          - ★ Academic integrity and ethics strongly followed.

Recap of previous class

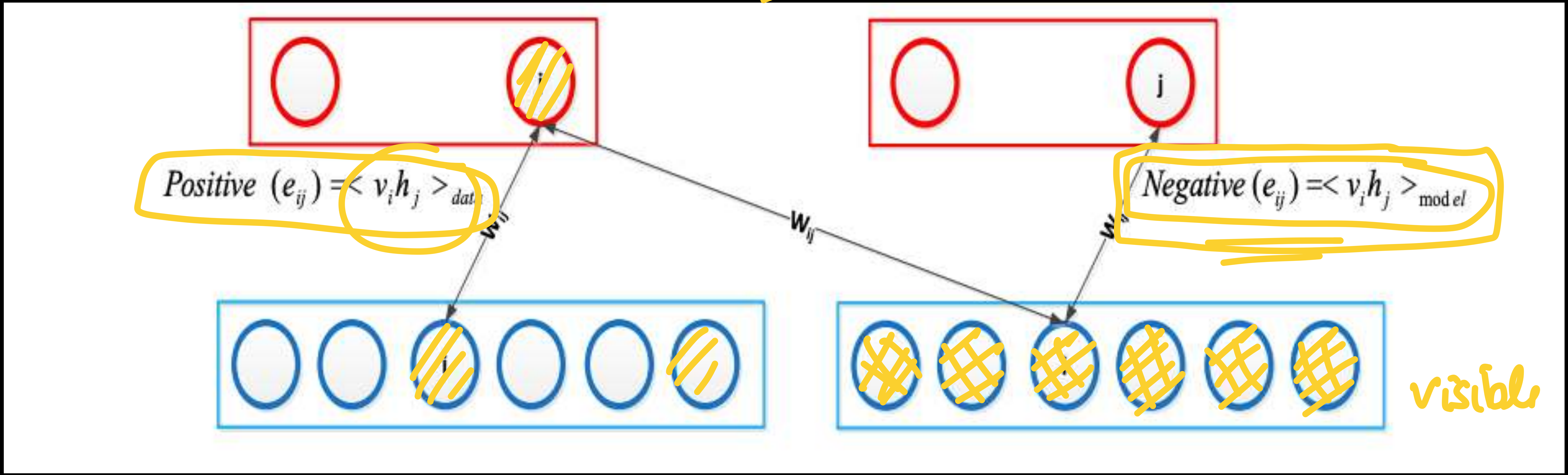
Restricted Boltzmann machines

- Properties
- Training
  - → Approximation

Contrastive Divergence

$v_i \quad p(h_j = 1 / v) = \sigma\left(\quad\right)$

$p(h_j = 1 / v) \ \forall j$



One-step contrastive divergence

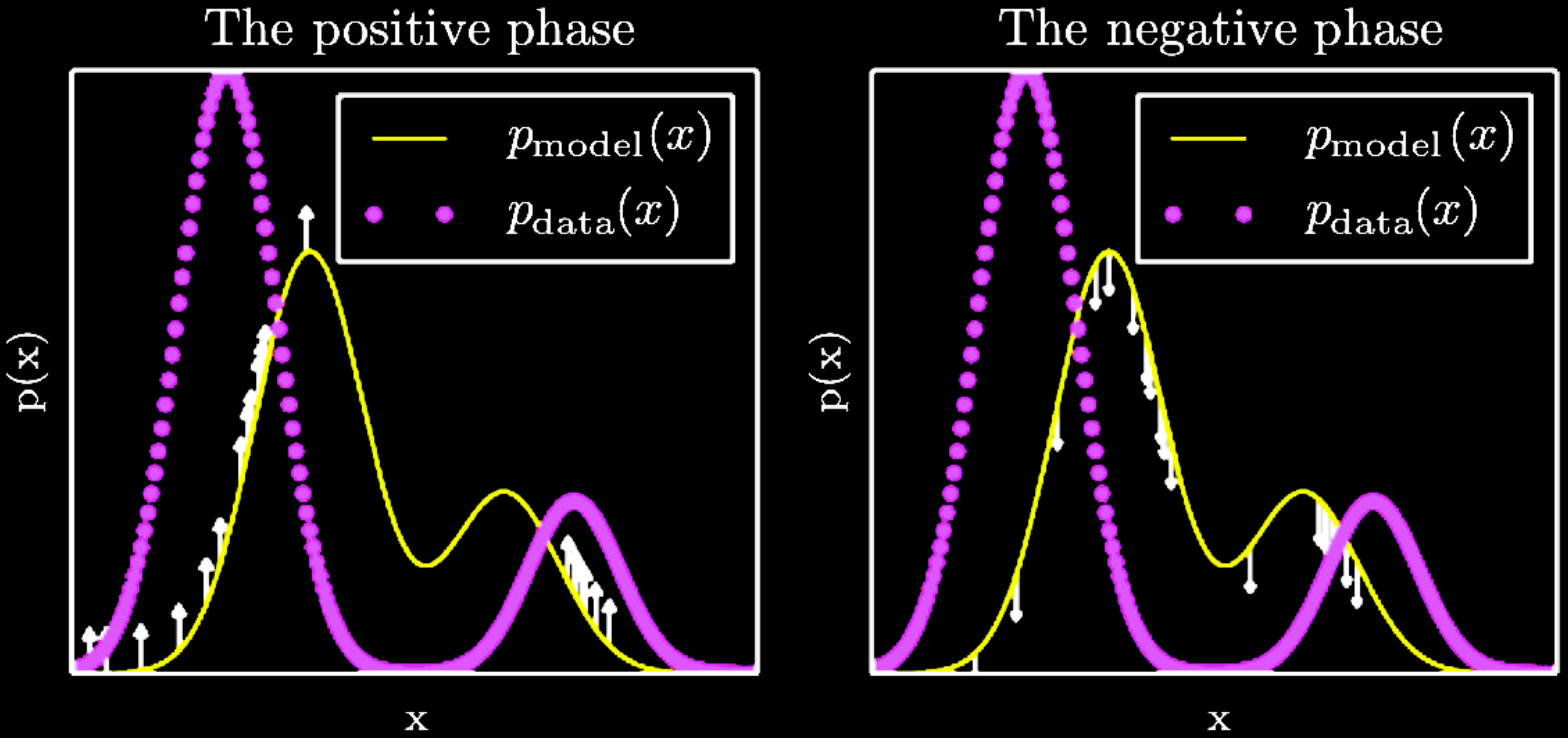
- Computing the gradient

$$\frac{\partial(p([\mathbf{v}\ \ \mathbf{h}], \Theta)}{\partial \mathbf{W}} \approx \frac{1}{N}\sum_{q=1}^{N} \mathbf{v}_q \mathbf{h}_q^T - \frac{1}{N}\sum_{q=1}^{N} \tilde{\mathbf{v}}_q \tilde{\mathbf{h}}_q^T$$

- Performing gradient ascent using the approximate gradient

$$\mathbf{\Theta}^{k+1} = \mathbf{\Theta}^{k} + \eta \frac{\partial log(p(\mathbf{x}, \Theta))}{\partial \mathbf{\Theta}}\bigg|_{\mathbf{\Theta}=\mathbf{\Theta}^k}$$

Positive phase and negative phase



Gaussian Bernoulli RBM

- For modeling real observations $\mathbf{v} \in \mathcal{R}^D$

- Define the energy function

$$E[\mathbf{v},\mathbf{h}] = \frac{1}{2}\underline{(\mathbf{v}-\mathbf{a})^T(\mathbf{v}-\mathbf{a})} - \mathbf{v}^T\mathbf{W}\mathbf{h} - \mathbf{b}^T\mathbf{h}$$

$$p([\mathbf{v},\mathbf{h}]) = \frac{e^{-E[\mathbf{v},\mathbf{h}]}}{Z}$$

✓

- The conditional distributions

Perform these derivations

$$p(\mathbf{v}|\mathbf{h}) = \mathcal{N}(\mathbf{W}\mathbf{h}+\mathbf{a},\mathbf{I})$$

$$p(h_j = 1|\mathbf{v}) = \sigma(\mathbf{v}^T\mathbf{W}^k_{:,j} + b_j)$$

Properties of GRBM

- d = 0

$$E(\mathbf{v}) = \frac{1}{2}(\mathbf{v} - \mathbf{a})^T(\mathbf{v} - \mathbf{a})$$

$$p(\mathbf{v}) = \frac{e^{-E(\mathbf{v})}}{Z}$$

- The marginal distribution is a Gaussian.

Properties of GRBM

- d = 1

$$E[\mathbf{v}, h] = \frac{1}{2}(\mathbf{v} - \mathbf{a})^T(\mathbf{v} - \mathbf{a}) - h\mathbf{v}^T\mathbf{w} - hb$$

$$p([\mathbf{v}, h]) = \frac{e^{-E[\mathbf{v},h]}}{Z}$$

$$p([\mathbf{v}, h = 0]) = \alpha\mathcal{N}(\mathbf{a}, \mathbf{I})$$

$$p([\mathbf{v}, h = 1]) = (1 - \alpha)\mathcal{N}(\mathbf{a} + \mathbf{w}, \mathbf{I})$$

- The marginal distribution is then

$$p(\mathbf{v}) = p([\mathbf{v}, h = 0]) + p([\mathbf{v}, h = 1])$$

✓ 2-mixture Gaussian

Properties of GRBM

$$\underline{h}_d = \begin{bmatrix} \underline{h}_{d-1} \\ h_d \end{bmatrix}$$

0 1

- For any general d dimensions

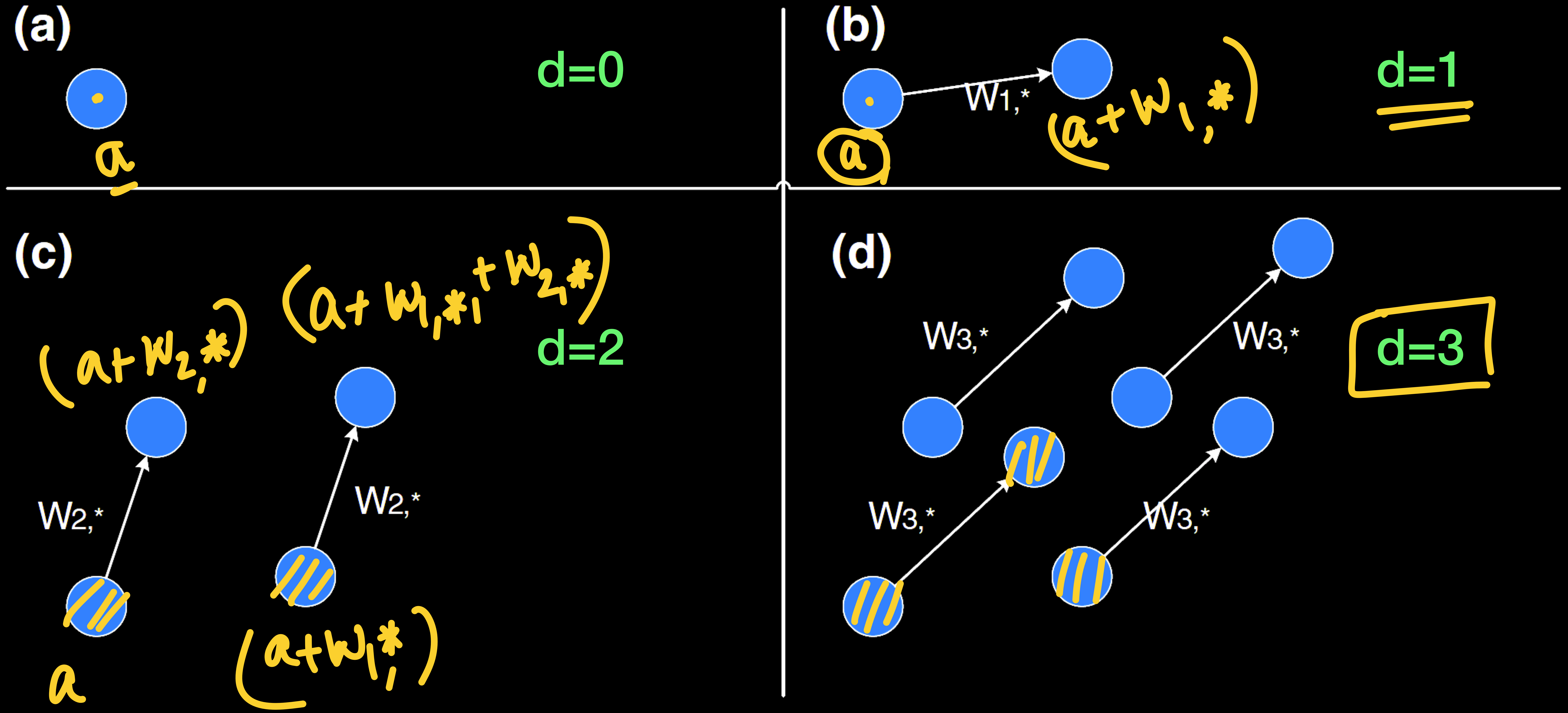
$$E[\mathbf{v}, \mathbf{h}_d] = E[\mathbf{v}, \mathbf{h}_{d-1}] + h_d\, \mathbf{v}^T \mathbf{W}_{:,d} + b_d h_d$$

$$p([\mathbf{v}, [\mathbf{h}_{d-1}, h_d = 0]]) = \alpha\, p([\mathbf{v}, \mathbf{h}_{d-1}])$$

$$p([\mathbf{v}, [\mathbf{h}_{d-1}, h_d = 1]]) = (1-\alpha)\, p([(\mathbf{v} + \mathbf{W}_{:,d}), \mathbf{h}_{d-1}])$$

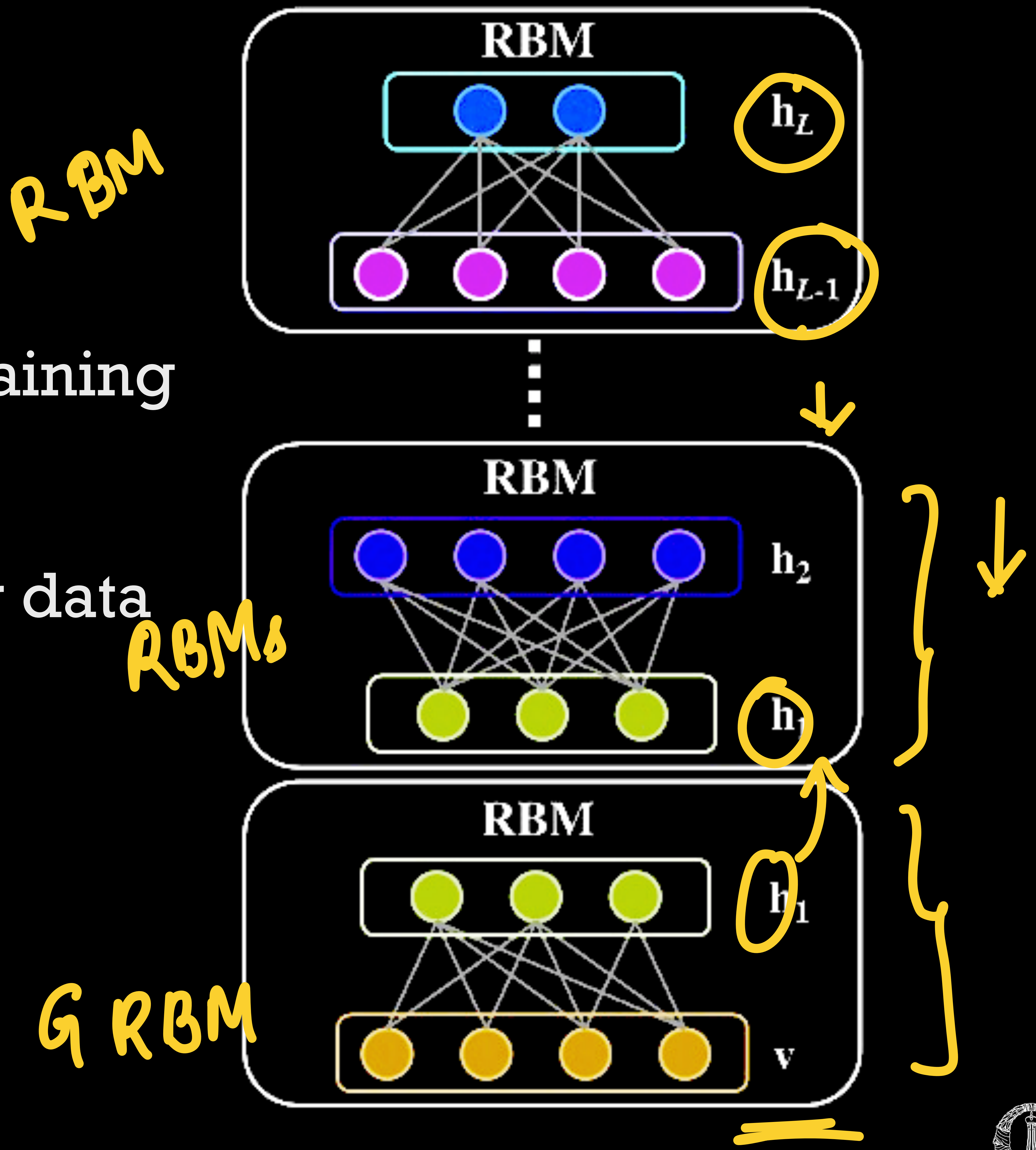
- For d=0, 1 Gaussian, d=1, 2-mix Gaussian, …
  - 2^d mixture Gaussian for any arbitrary d.

GRBMs and GMMs



Deep Belief Networks (DBN)

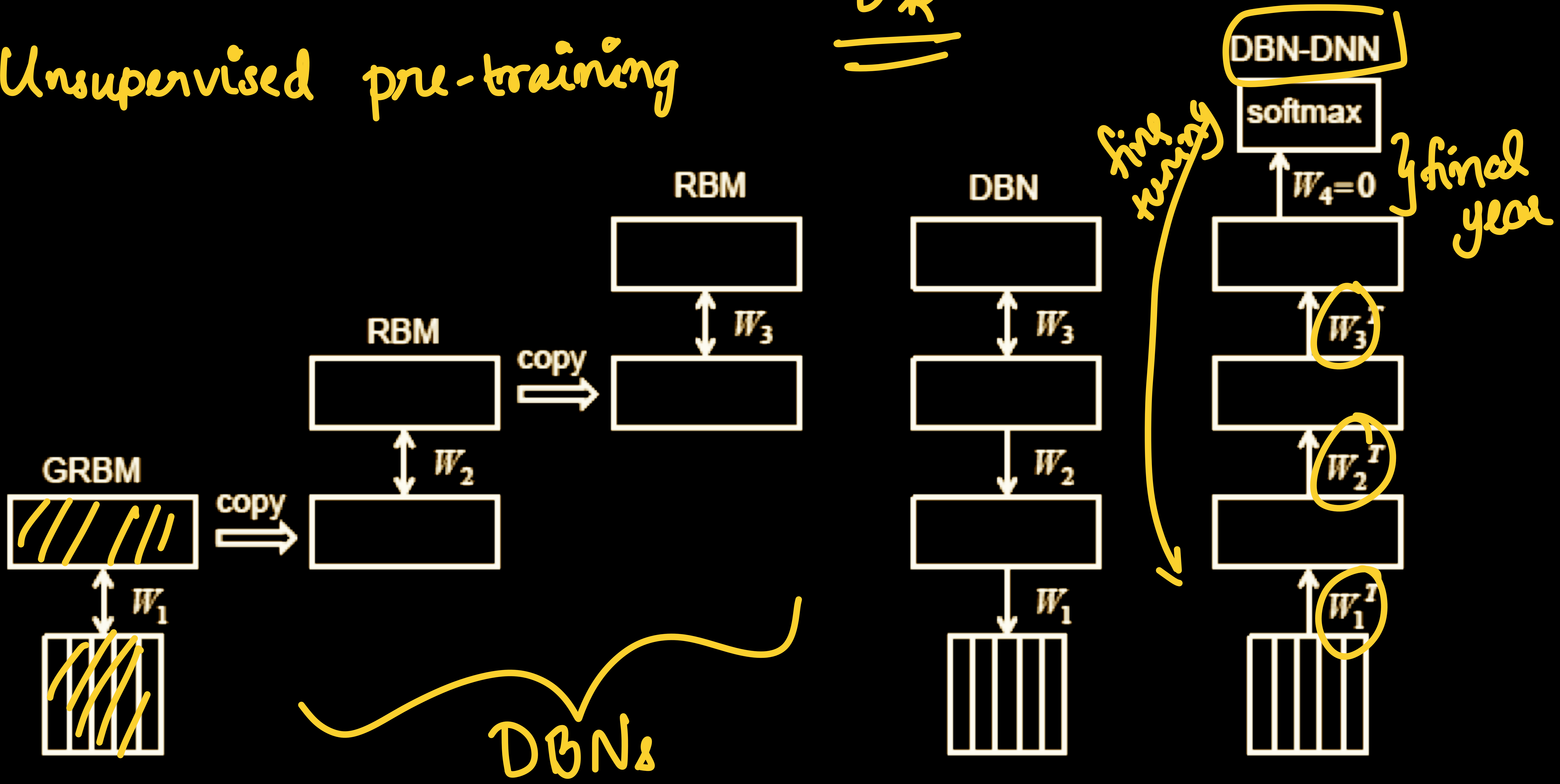
- Stacking RBMs in a disjoint fashion
  - ✓ Layer-wise training for each RBM.
    - ◉ Weights are frozen each layer before training the next layer.
  - ✓ Ancestral sampling can be performed for data generation
    - ★ Lossy sample generation due to accumulation of errors.
- Most common use - pre-training of DNNs.



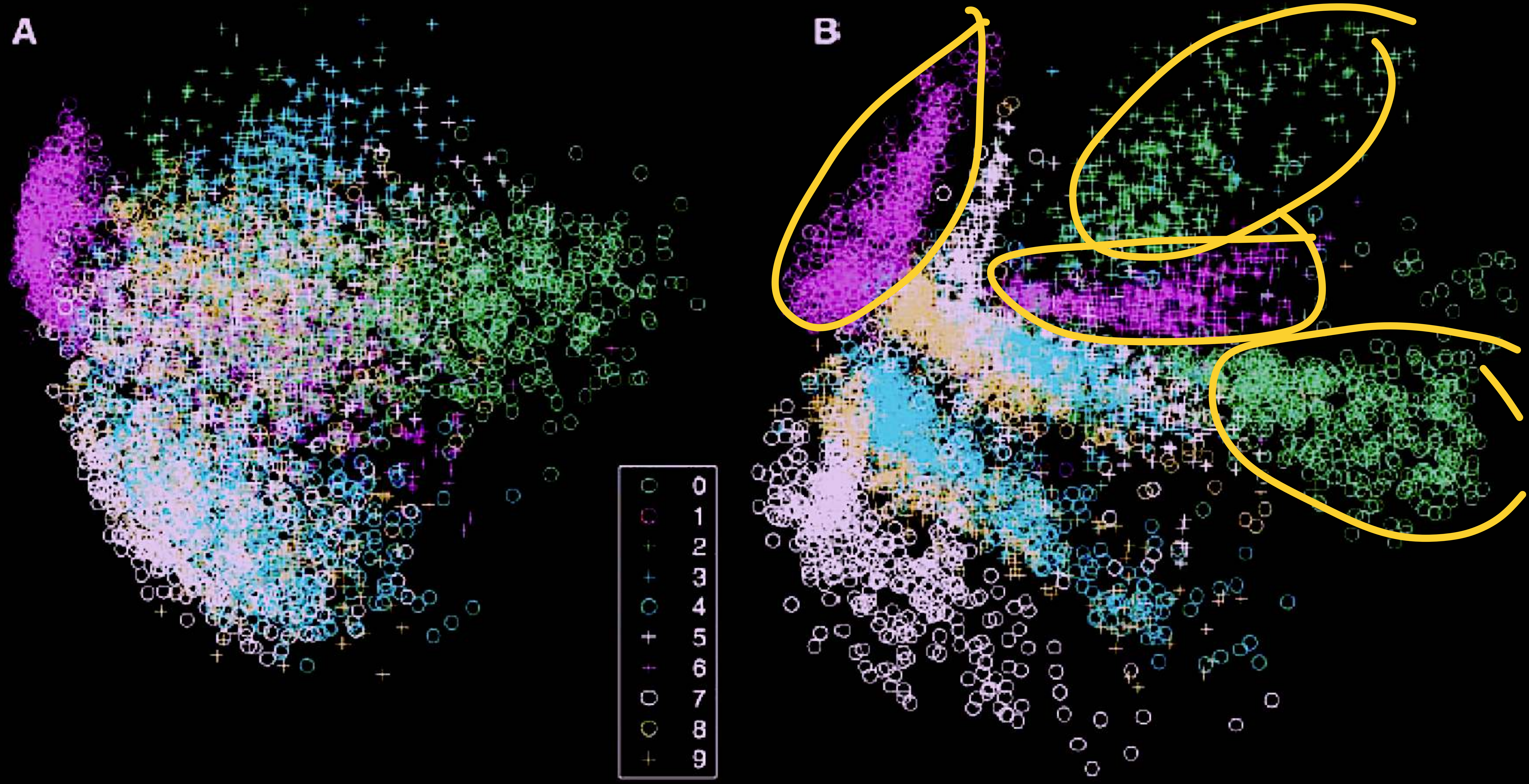
DBNs for initialization

Unsupervised pre-training

5k



DBNs for visualization



PCA RBM

t-SNE, PCA and DBNs

Unsupervised ✓

Unseen data [PCA, DBN]

Neighborhood [t-SNE]

Hierarchy based [DBNs]

↑ binary

Distribution based [t-SNE, DBNs]

Data generation ? [DBNs]

Initilization

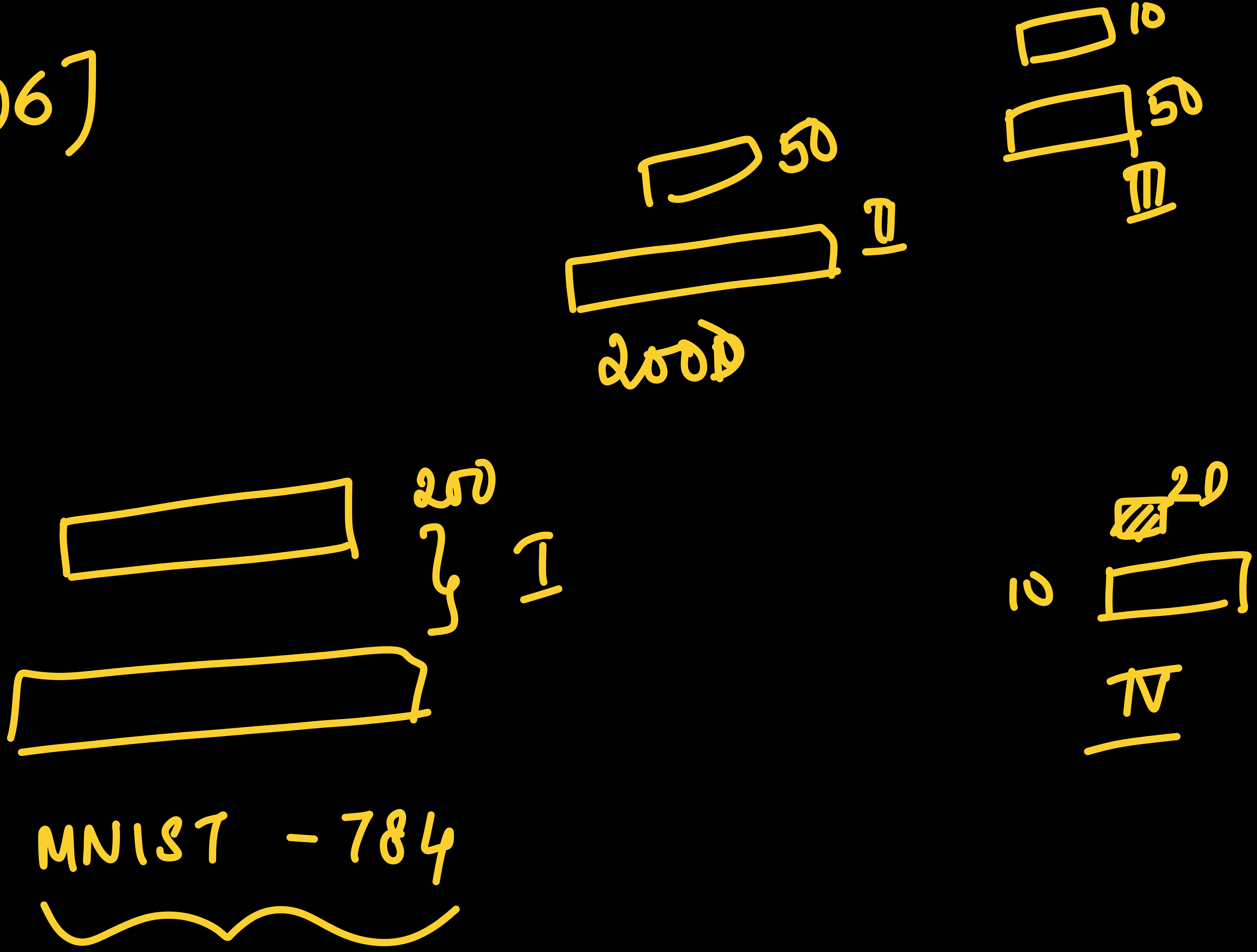
* Generative modeling

* Expected Hidden units conditioned on visible

$$- \sigma \left(\qquad \right)$$

Affine + sigmoid [DNN]

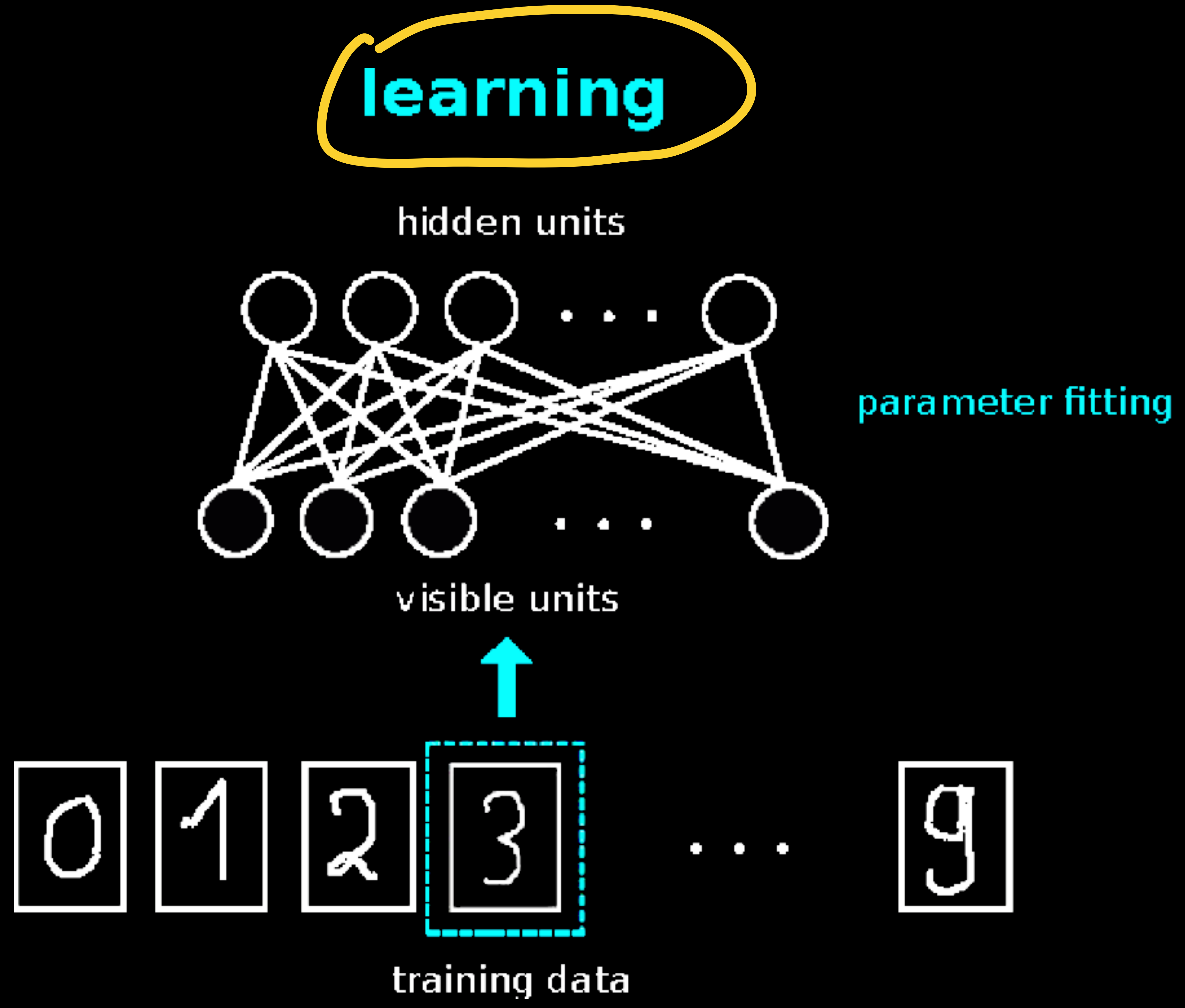
[Hinton, 2006]

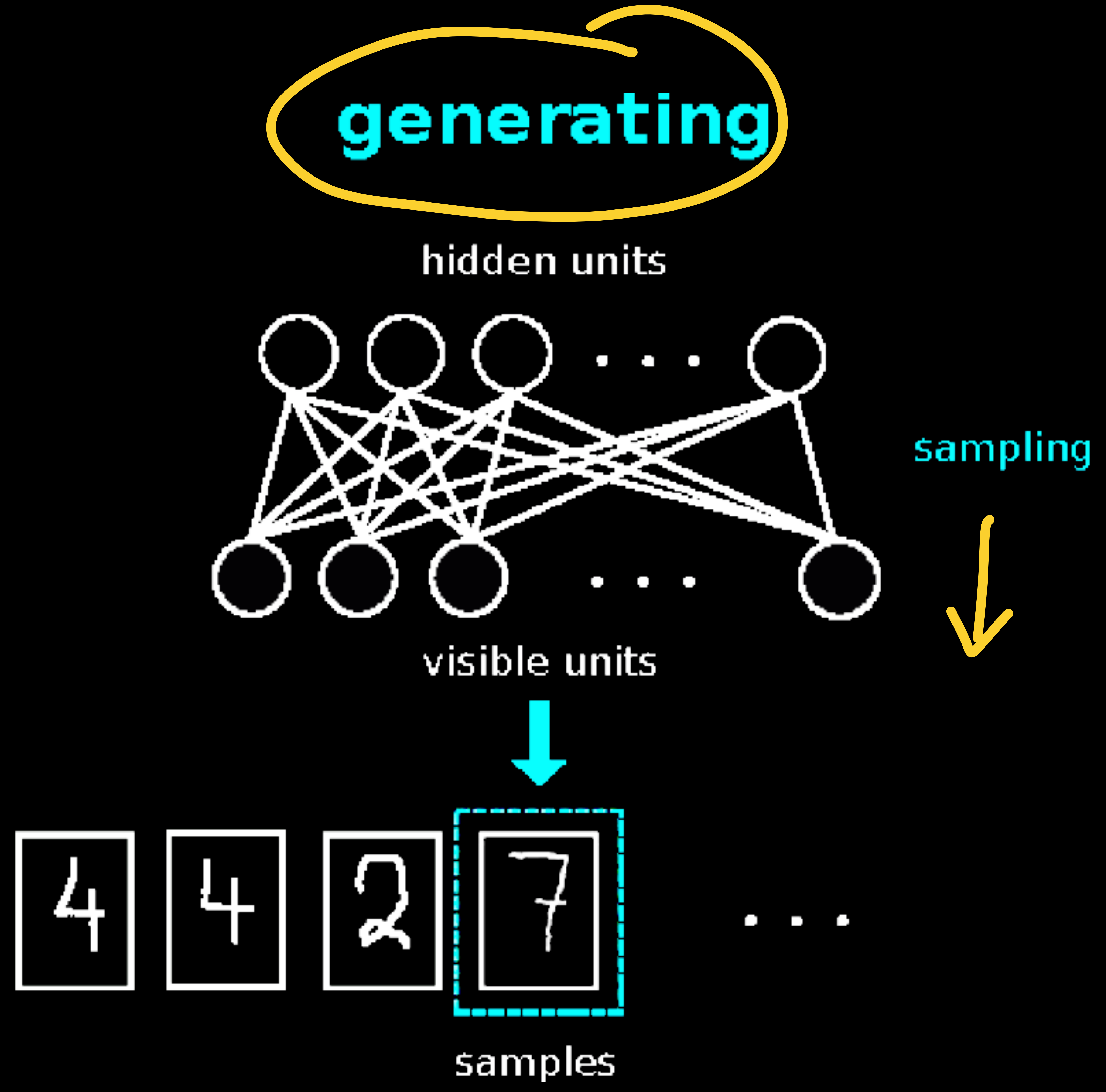


More reading

Salakhutdinov, Ruslan, Andriy Mnih, and Geoffrey Hinton. "Restricted Boltzmann machines for collaborative filtering." *Proceedings of the 24th international conference on Machine learning*. 2007.

Data generation using RBMs





Data generation using RBMs

0

255

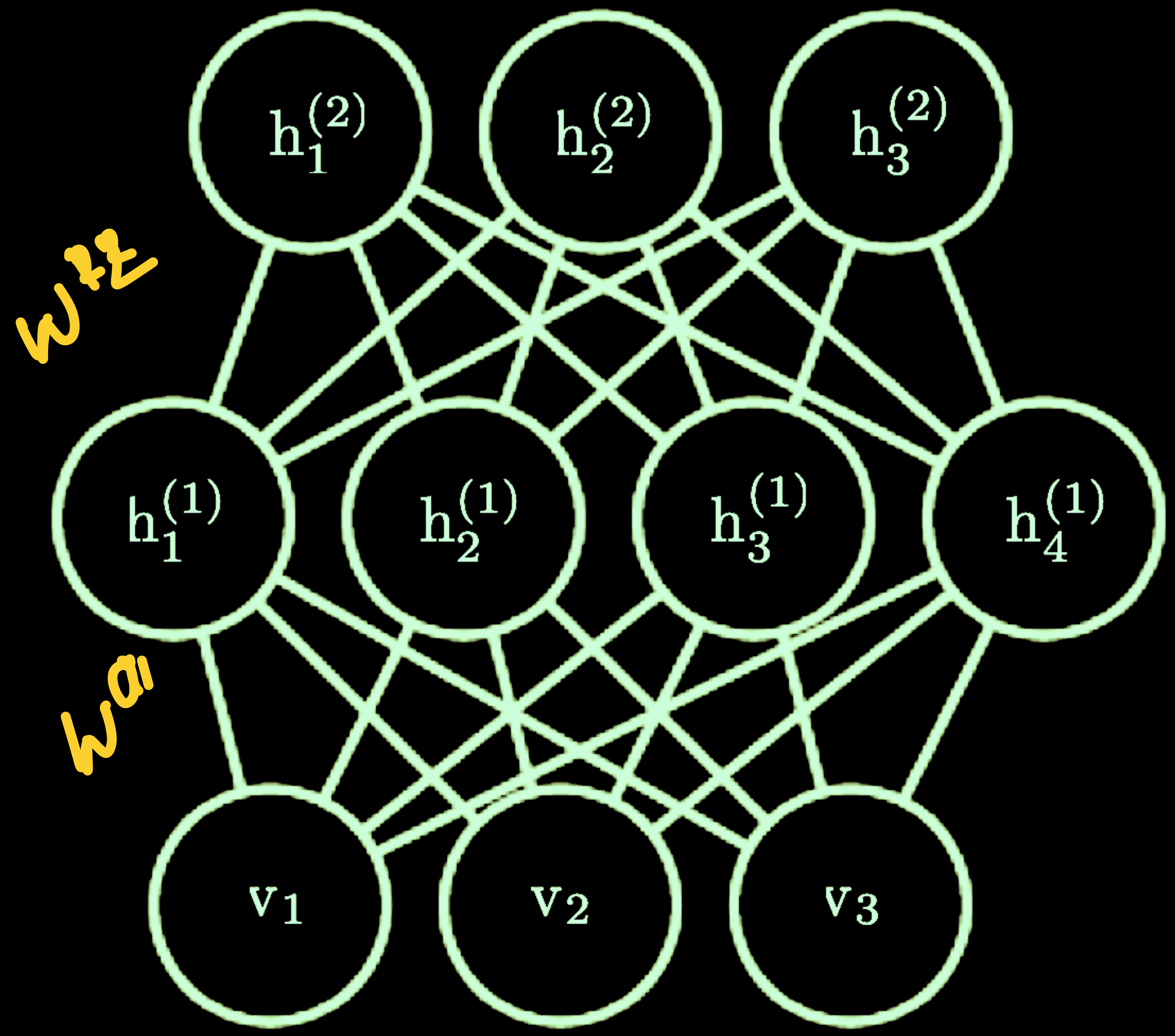
28x28

8

Source — https://lme.tf.fau.de/lecture-notes/lecture-notes-in-deep-learning-unsupervised-learning-part-1/

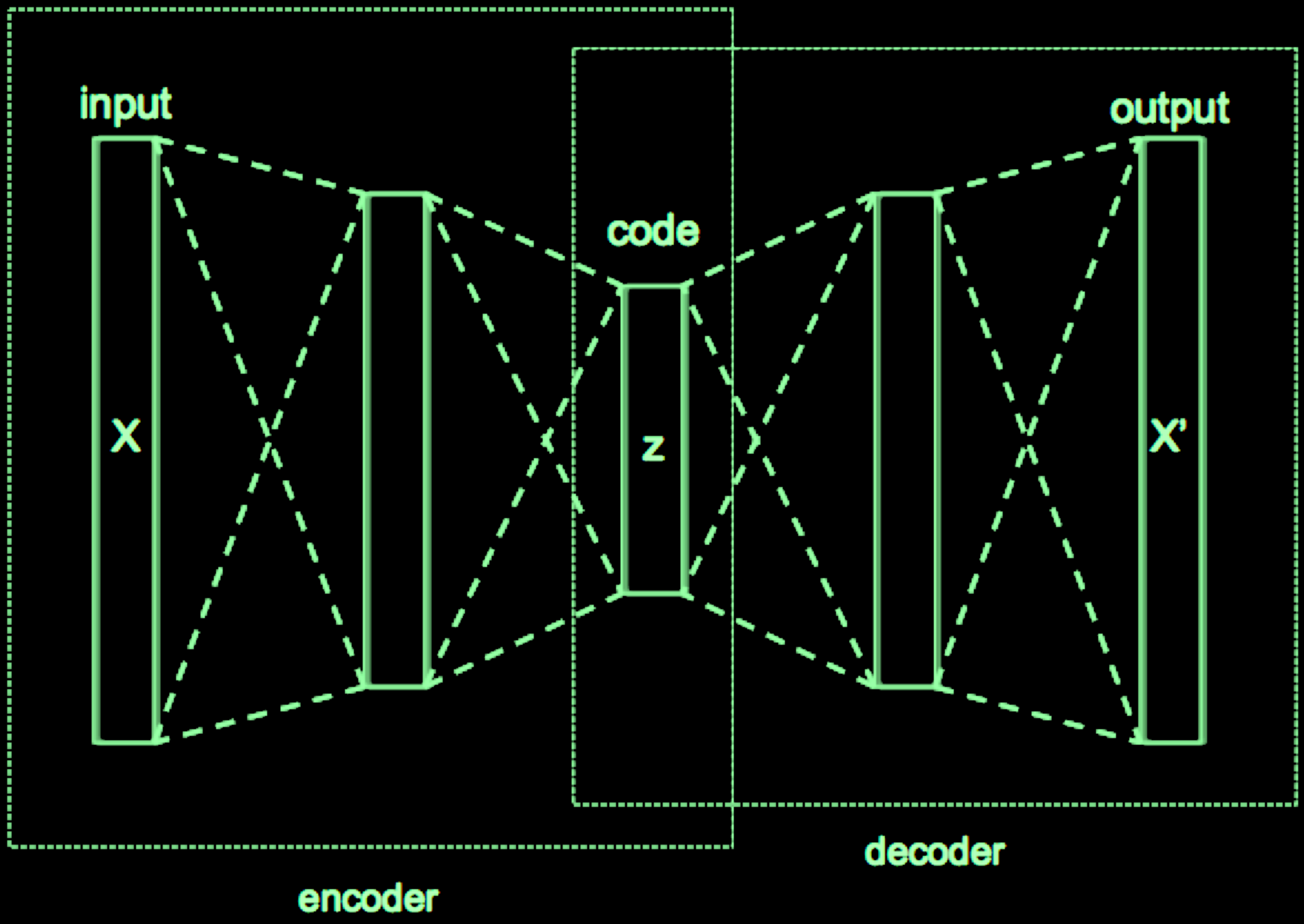
Deep Boltzmann machine (DBMs)

- Deep layers of connections with RBM structure.
  - Joint energy function.
  - Undirected graph
- Training and inference are more involved.



Autoencoders

- Encoder decoder
  - Latent representations capture a lower dimensional embedding of the data.
    - ✓ can be feedforward or convolutional layers.
- Model training
  - Using a reconstruction loss.



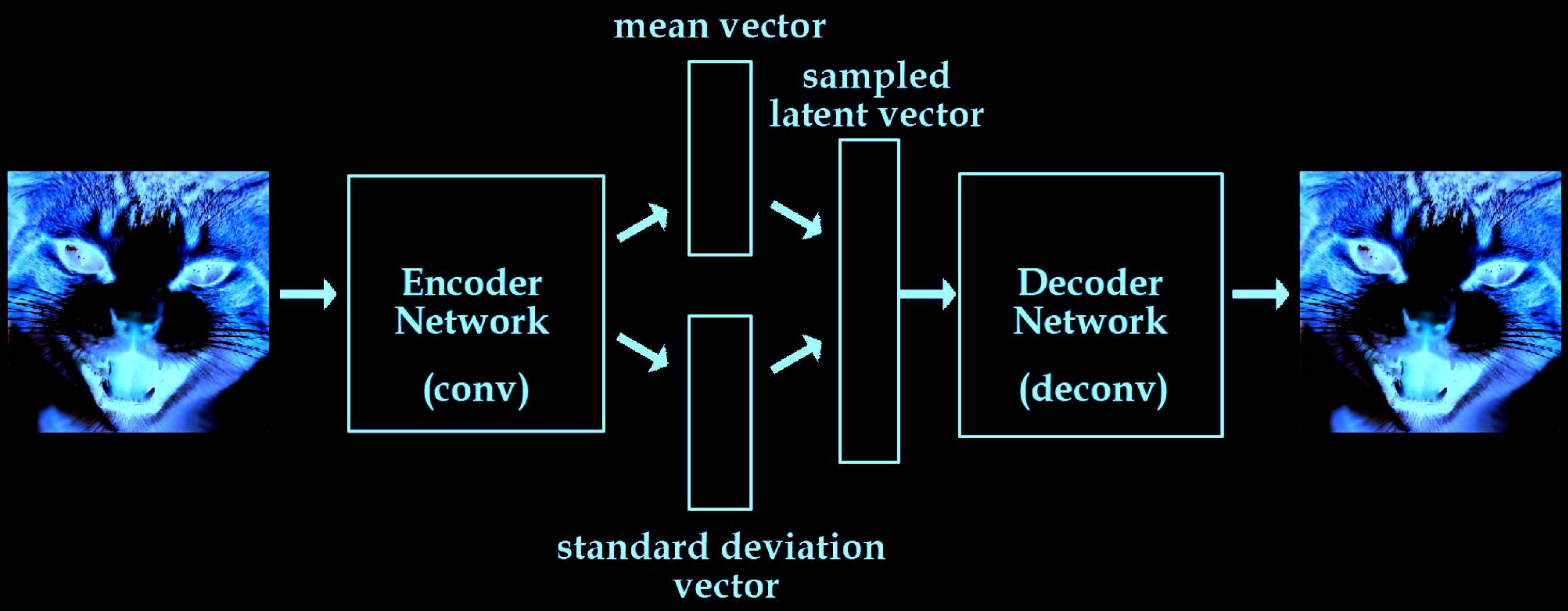
Variational auto encoders

VAEs v/s RBMs

(VAE) latent variables → real valued

(RBM) latent → binary

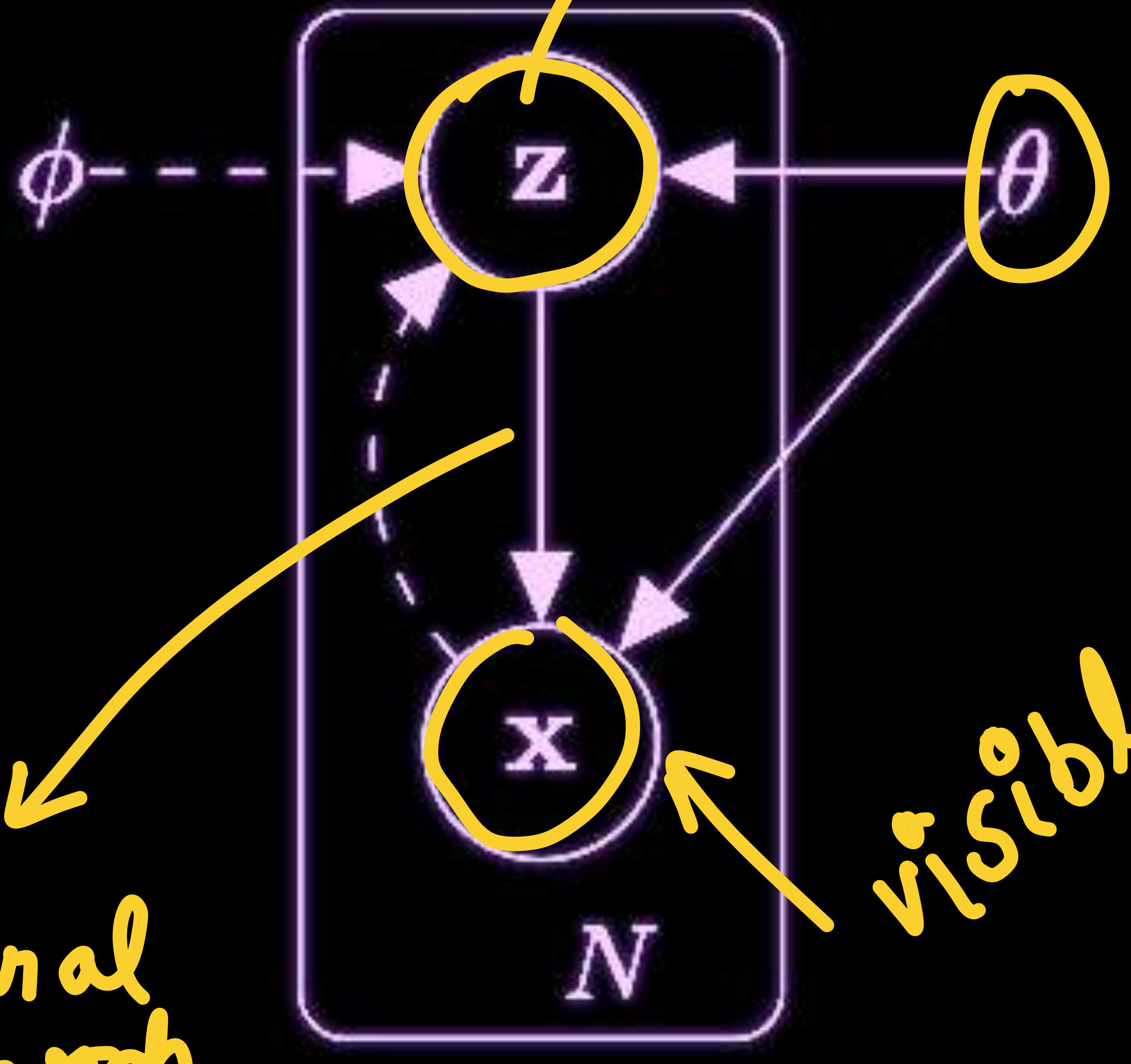
- ✴️ Sampling from the latent space.
  - ➡️ Making Gaussian assumptions at the latent layer



Variational auto encoders

- The data $\mathbf{x}$ and latent variable $\mathbf{z}$
- The forward model
  - ✓ Sample the latent variable $p_\theta(z)$
  - ✓ Sample the data given the latent $\underbrace{p_\theta(\mathbf{x}|\mathbf{z})}$
- The marginal distribution $p_\theta(x) = \int \underbrace{p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})}d\mathbf{z}$
  - ✓ maybe intractable
- The posterior distribution $\underbrace{p_\theta(\mathbf{z}|\mathbf{x})}$
  - ➡️ may also be intractable

backward



Variational auto encoders

Variational lower bound

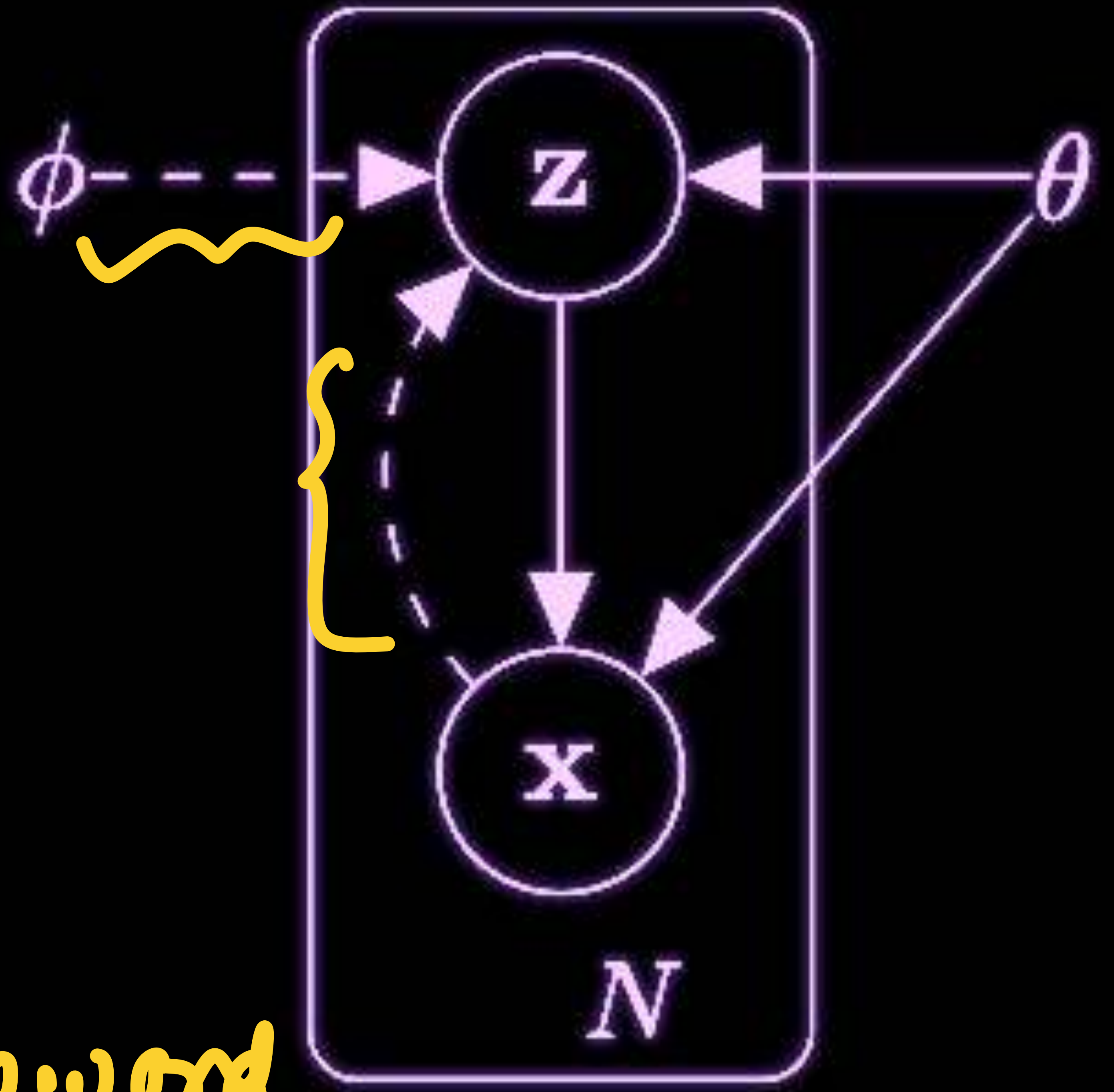
- Approximate the posterior

$$\boxed{q_\phi(\mathbf{z}|\mathbf{x}) \sim p_\theta(\mathbf{z}|\mathbf{x})}$$

$$\boxed{p_\theta(z|x)}$$

approx.

- Using variational lower bound.



A neural network θ → forward model

$z \longrightarrow x$

A second neural network ϕ → backward model

$x \longrightarrow z$

Variational lower bound

$$
\begin{aligned}
k = \underline{\log p_\theta(\mathbf{x})} &= \log \int \underbrace{p_\theta(\mathbf{x},\mathbf{z})d\mathbf{z}} \\
&= \log \int \underbrace{p_\theta(\mathbf{x},\mathbf{z})\frac{q_\phi(\mathbf{z}|\mathbf{x})}{q_\phi(\mathbf{z}|\mathbf{x})}}d\mathbf{z} \\
&= \log_w \left(\tilde{\mathbb{E}}_q\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]\right) \\
&\underbrace{\geq \mathbb{E}_q}\left(\log\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]\right) \\
&= \cancel{\mathbb{E}_q\left(\log\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]\right)} \\
&= \underbrace{\mathbb{E}_q\left(\log\left[p_\theta(\mathbf{x},\mathbf{z})\right]\right)} + \underbrace{H_q(\mathbf{z}|\mathbf{x})}
\end{aligned}
$$

$$\frac{q_\phi(z/x)}{q_\phi(z/x)}$$

Jensen's inequality

$$-\int_z \underbrace{\log q_\phi \cdot q_\phi\, dz}_{H_q}$$

Variational lower bound

$$
\begin{aligned}
KL\left[q(Z)\|p(Z|X)\right] &= \int_Z q(Z)\log\frac{q(Z)}{p(Z|X)} \\
&= -\int_Z q(Z)\log\frac{p(Z|X)}{q(Z)} \\
&= -\left(\int_Z q(Z)\log\frac{p(X,Z)}{q(Z)} - \int_Z q(Z)\log p(X)\right) \\
&= -\int_Z q(Z)\log\frac{p(X,Z)}{q(Z)} + \log p(X)\int_Z q(Z) \\
&= -L + \log p(X)
\end{aligned}
$$